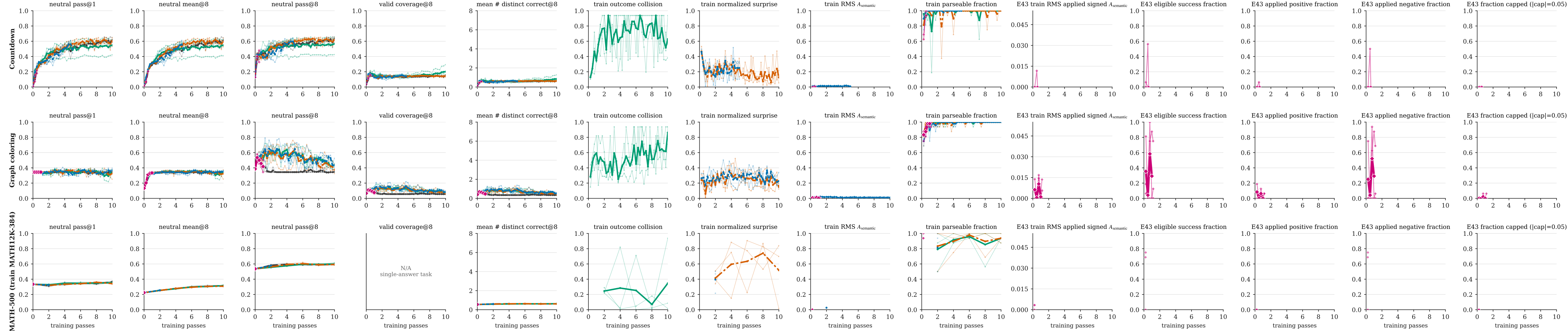


— Matched Dr.GRPO
 — Semantic collision entropy
 - - Predictive semantic Shannon entropy
 - - Separately-centered semantic Shannon advantage (E41)
 ◆ success-conditioned signed Shannon advantage (E43)

E37/E38/E39/E41/E43 free-form semantic diversity — matched head-to-head



Countdown/Graph use identical neutral K=8 evaluation prompts. MATH-500 uses one fixed K=8 draw (seed 390100) every two passes; coverage@8 is N/A for this single-answer task, and distinct@8 counts distinct normalized correct representations, not reasoning paths. Collision, predictive Shannon surprise, E41 separately-centered semantic advantage, and E43 success-conditioned signed Shannon advantage are on-policy diagnostics. E43 learns semantic support from verified successes and retains bounded positive and negative pressure on eligible successes; its applied RMS, eligibility, signed fractions, and cap-hit fraction are shown. Thin lines are seeds and heavy lines require all three paired seeds.